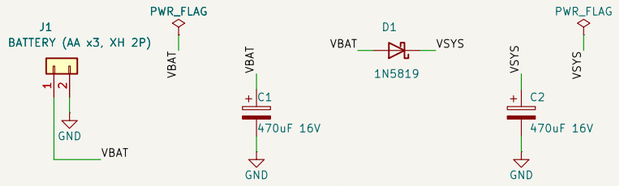
Linetracer2 Lite rev.L3 : XIAO ESP32C6 on pin headers, 3 x LBR-127HLD, 1 button, 1 buzzer, 6 RGB LEDs

Power: AA x3 (alkaline 4.5V recommended / NiMH 3.6V) -> VBAT (motors) ; VBAT -> D1 Schottky -> VSYS -> XIAO 5V pin. USB 5V can not back-feed the batteries. Never use 4 cells (the XIAO's 5V input clamps above about 6 V).

1. POWER



Insert the plug first and solder the header so that the RED wire is on '+'

XIAO 5V pin = USB VBUS: an external supply needs a diode (D1), otherwise USB 5V charges the AA cells.

Battery voltage is NOT measured: the XIAO has 3 ADC pins only (D0..D2) and the sensors use all of them.

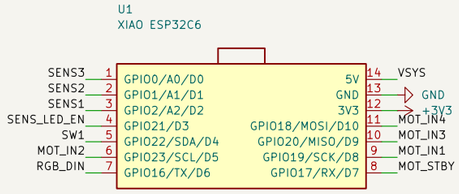
Inside the XIAO: 5V pin -> Schottky -> buck regulator -> 3V3 (3V3 stays up while VBAT > about 4 V).



MECHANICAL: H1-H4 = gearbox frame M3 (x30/x70, y62/y96), H5 = front ball caster / skid, snap-in (x50, y11: behind the centre sensor)

H1 M3 H2 M3 H3 M3 H4 M3 H5 SKID M3

2. MCU Seeed Studio XIAO ESP32C6 (on two 1x7 pin headers)



GPIO15 = yellow user LED on the XIAO (on = 0), a 2nd indicator for free. BOOT / RESET buttons are on the XIAO.

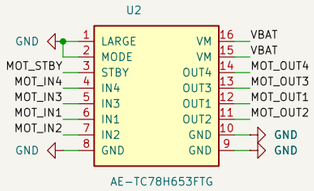
Reset state (ESP32-C6 datasheet): GPIO18-23 weak pull-up during reset, then floating;

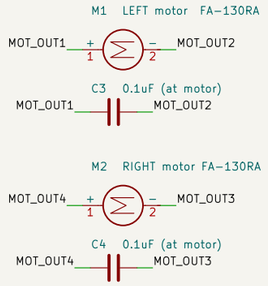
GPIO16 (TX) / GPIO17 (RX) pull-up after reset + boot log / UART REPL -> no button and no motor input there.

On pin headers the XIAO sits 2.5 mm above the board: its bare bottom test pads can not touch the tracks.

Never solder a Li-ion cell to the XIAO's BAT pads on this robot (AA cells are on the 5V pin).

3. MOTOR DRIVER (Akizuki AE-TC78H653FTG module, DIP16)





LARGE=L : small mode (2 motors, 2A/ch) MODE=L : IN/IN control

IN1/IN2 -> OUT1/OUT2 = LEFT motor, IN3/IN4 -> OUT3/OUT4 = RIGHT motor (+ = OUT4)

H/L fwd, L/H rev, H/H brake, L/L coast. Over-current (ISD) latches OFF:

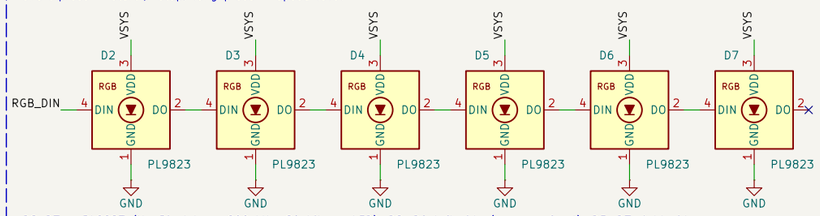
toggle STBY L->H to recover (done by the library).

4. BUTTON / BUZZER / RGB LEDs



internal pull-up, pressed = 0. R10: if START is pressed while D4 beeps, only 3.3 mA flows

short press = start / stop, long press = speed level



D2..D7 = PL9823 (NeoPixel type, 800 kHz, 24 bit per LED): D2-D4 left side (rear -> front), D5-D7 right side (front -> rear). Data from D6 (TX: the boot log may flash them once). 6 x white = 0.36 A: the library limits it.

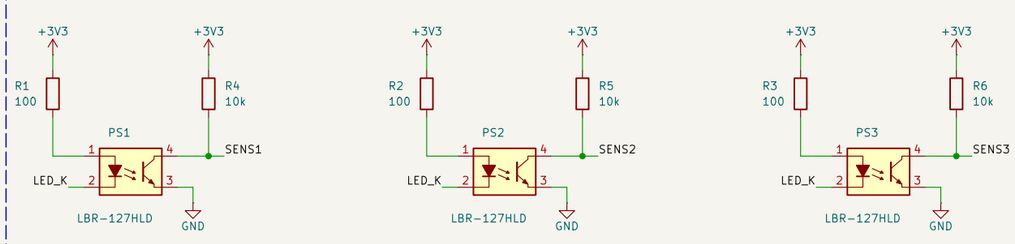


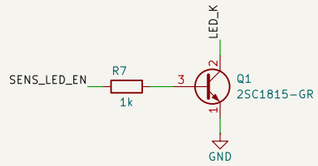
The buzzer shares D4 with the START button (no free XIAO pin). A piezo passes no DC, so reading the button (input + pull-up) works; to beep, D4 becomes a PWM output - it also sounds while START is held.

5. LINE SENSORS 3 x LBR-127HLD (mount on the BOTTOM side, lens to the floor), 12 mm apart

IR LED: (3.3V - 1.2V - 0.1V) / 100 ohm = about 20 mA each. Q1 switches all LEDs (ambient light cancel). Photo transistor: 10k pull-up -> white = low voltage, black = high voltage

The body is 5.6 mm tall: the front of the robot is lifted by the 7.5 mm ball caster or skid (lens about 2.25 mm above the floor)





Q1: all IR LEDs on when SENS_LED_EN = 1 (about 60 mA)

PS1 = LEFT -> D2 (GPIO2/ADC), PS2 = CENTRE -> D1 (GPIO1/ADC), PS3 = RIGHT -> D0 (GPIO0/ADC) (seen from above, robot moving up the page)

6. TEST PADS (multimeter, black probe on GND)



VBAT (D1 anode / J1 +) 3.6-4.8 V, VSYS = VBAT - 0.3 V (USB: 5 V), 3V3 = 3.3 V.

S1-S3: white low, black about 3 V.

IR (LED_K): about 0.1 V while the IR LEDs are on (robot.sensors.leds(True)),

about 2 V when off.

Design notes: lite/README.md
XIAO ESP32C6 + Akizuki AE-TC78H653FTG module + 3 x LBR-127HLD
All parts through-hole. XIAO on two 1x7 pin headers
EDTC

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Title: Linetracer2 Lite rev.L3 - XIAO ESP32C6 kids line tracer

Size: A3	Date: 2026-10-07	**Rev: L3**
KiCad E.D.A. 10.0.7		Id: 1/1